

30 CHAPTERWISE IMPORTANT QUESTIONS

Science (086) | CBSE 2026-27

Electricity

CBSE Science 2026-27 | Board-examinable

Exactly 20 numericals followed by 10 theory questions, with step-based answers for Class X Physics revision.

Question 1

A 6 ohm resistor is connected to 12 V. Find current and power.

Answer:

$$I=V/R=12/6=2 \text{ A. } P=VI=12 \times 2=24 \text{ W.}$$

Question 2

A 10 ohm resistor carries 0.5 A. Find potential difference and power.

Answer:

$$V=IR=0.5 \times 10=5 \text{ V. } P=VI=5 \times 0.5=2.5 \text{ W.}$$

Question 3

Find equivalent resistance of 4 ohm and 6 ohm resistors in series.

Answer:

$$R=4+6=10 \text{ ohm.}$$

Question 4

Find equivalent resistance of 4 ohm and 6 ohm resistors in parallel.

Answer:

$$1/R=1/4+1/6=5/12, \text{ so } R=12/5=2.4 \text{ ohm.}$$

Question 5

Three resistors 2 ohm, 3 ohm and 5 ohm are in series across 20 V. Find circuit current.

Answer:

$$R=2+3+5=10 \text{ ohm. } I=V/R=20/10=2 \text{ A.}$$

Question 6

Two 6 ohm resistors are in parallel across 12 V. Find equivalent resistance and total current.

Answer:

$$R=3 \text{ ohm. } I=V/R=12/3=4 \text{ A.}$$

Question 7

A 100 W bulb runs for 5 hours. Find energy in kWh.

Answer:

$$\text{Power}=0.1 \text{ kW. Energy}=0.1 \times 5=0.5 \text{ kWh.}$$

Question 8

A 1.5 kW heater works for 2 hours daily for 30 days. Find units consumed.

Answer:

$$\text{Energy}=1.5 \times 2 \times 30=90 \text{ kWh, or 90 units.}$$

Electricity - Continued

Question 9

An appliance consumes 2 kWh in 4 hours. Find its power.

Answer:

$$P = E/t = 2/4 = 0.5 \text{ kW} = 500 \text{ W}.$$

Question 10

A current of 3 A flows through a 4 ohm resistor for 2 minutes. Find heat produced.

Answer:

$$H = I^2 R t = 3^2 \times 4 \times 120 = 4320 \text{ J}.$$

Question 11

A 60 W bulb is used for 10 hours. Find energy in joules.

Answer:

$$E = P t = 60 \times (10 \times 3600) = 2,160,000 \text{ J}.$$

Question 12

A wire has $R=5$ ohm and $I=2$ A. Find heat in 30 seconds.

Answer:

$$H = I^2 R t = 2^2 \times 5 \times 30 = 600 \text{ J}.$$

Question 13

A resistor takes 2 A from a 220 V supply. Find resistance and power.

Answer:

$$R = V/I = 220/2 = 110 \text{ ohm}. \quad P = VI = 440 \text{ W}.$$

Question 14

A 220 V, 1100 W heater is connected to its rated supply. Find current.

Answer:

$$I = P/V = 1100/220 = 5 \text{ A}.$$

Question 15

A wire has resistance 8 ohm. If its length is doubled with same area, find new resistance.

Answer:

R is proportional to length. New $R = 2 \times 8 = 16 \text{ ohm}$.

Question 16

A wire has resistance 12 ohm. If cross-sectional area is doubled with same length, find new resistance.

Answer:

R is inversely proportional to area. New $R = 12/2 = 6 \text{ ohm}$.

Question 17

Resistors 3 ohm and 6 ohm are in parallel, then in series with 2 ohm. Find total resistance.

Answer:

Parallel part: $R = 2 \text{ ohm}$. Total $R = 2 + 2 = 4 \text{ ohm}$.

Electricity - Continued

Question 18

A 4 ohm resistor has 8 V across it. Find current, power and heat in 10 s.

Answer:

$I=8/4=2$ A; $P=VI=16$ W; $H=Pt=160$ J.

Question 19

A household uses 4 bulbs of 15 W for 6 hours. Find energy in kWh.

Answer:

Total power=60 W=0.06 kW. Energy=0.06 x 6=0.36 kWh.

Question 20

Electricity costs Rs 6 per unit. Find cost of 75 kWh.

Answer:

Cost=75 x 6=Rs 450.

Question 21

State Ohm's law.

Answer:

At constant temperature, current through a conductor is directly proportional to potential difference across it: $V=IR$.

Question 22

Why are household appliances connected in parallel?

Answer:

Each appliance gets the full supply voltage and can work independently of the others.

Question 23

Differentiate series and parallel combination of resistors.

Answer:

In series, current is same and resistances add. In parallel, voltage is same and reciprocal resistances add.

Question 24

What is the SI unit of resistance?

Answer:

Ohm. One ohm is the resistance when 1 V produces 1 A current.

Question 25

State Joule's law of heating.

Answer:

Heat $H=I^2Rt$. It increases with square of current, resistance and time.

Question 26

Why is a fuse connected in the live wire?

Answer:

It melts during excessive current and breaks the live connection, reducing shock and fire risk.

Electricity - Continued

Question 27

What does the commercial unit of electrical energy mean?

Answer:

One unit equals one kilowatt-hour, the energy used by 1 kW in 1 hour; it equals 3.6×10^6 J.

Question 28

How does resistance depend on a conductor's dimensions?

Answer:

R is proportional to length, inversely proportional to area and depends on material and temperature.

Question 29

Why does a bulb filament have high resistance?

Answer:

A high resistance filament converts electrical energy into heat and light effectively without melting easily.

Question 30

What is electric power?

Answer:

It is the rate at which electrical energy is used or converted. $P=VI$ and its SI unit is watt.

Sources: CBSE Science curriculum 2026-27; CBSE Class X sample papers and marking schemes 2026-27; NCERT textbooks.

ONE SHOT MASTERY